

Sahilkumar Prasad

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Professional Summary

Backend Engineer with **3+ years of experience** designing scalable backend systems using **Java and Spring Boot**. Experienced in building microservices, REST APIs, and distributed systems. Strong expertise in **Kafka-based event-driven architecture, Redis caching, and database optimization**. Passionate about building reliable, high-performance backend platforms.

Technical Skills

Languages: Java, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs

Architecture: Microservices, Distributed Systems, Event-Driven Architecture, System Design (LLD, HLD)

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Messaging Systems: Apache Kafka

Testing: JUnit, Mockito

Tools: Docker, Git, GitHub Actions, CI/CD

Professional Experience

Software Development Engineer

May 2023 – Present

Crescent Engineering and Consulting, Navi Mumbai

- Architected and deployed **8+ Spring Boot microservices** supporting core workflows and handling **500K+ requests/day** with **99.9% uptime**.
- Reduced **API latency by 45%** by implementing Redis caching and asynchronous request processing.
- Improved **database query performance by 50%** through indexing strategies and query optimization.
- Migrated tightly coupled services into a **Kafka-based event-driven architecture** enabling asynchronous processing across services.
- Implemented secure authentication and authorization using **Spring Security, JWT, and RBAC**.
- Improved production reliability by introducing structured logging, retry mechanisms, and standardized error handling.

Key Projects

Distributed Notification System

- Designed a scalable notification platform supporting email, SMS, and in-app notifications using **Kafka-based event-driven architecture**.
- Implemented asynchronous consumers to process notification events reliably.
- Built retry mechanisms and **dead-letter queues** for failed notifications.
- Implemented Redis-based rate limiting to prevent notification spam.

Distributed Rate Limiter

- Designed a distributed rate limiter using **Spring Boot and Redis** based on the Token Bucket algorithm.
- Achieved throughput of **50K+ requests per minute** with minimal latency overhead.
- Ensured cross-node consistency using atomic Redis operations.

Distributed URL Shortener

- Built a high-performance URL shortener using **Spring Boot, PostgreSQL, Redis, and Kafka**.
- Reduced read latency by **80%** using Redis caching for frequently accessed URLs.
- Implemented asynchronous analytics processing using Kafka.

Education

University of Mumbai

2020 – 2023

Bachelor of Science in Computer Science

CGPA: **9.57 / 10**

Achievements

- Solved **300+ Data Structures and Algorithms problems** on coding platforms including LeetCode.
- Designed backend systems capable of handling **500K+ production requests daily**.
- Built distributed backend projects demonstrating caching, rate limiting, and event-driven architecture.